Assignment 5

Metoder i Kunstig Intelligens

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February 27, 2026

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1 Modeling the Decision Problem

I have chosen to build a decision support system for a real-world decision: whether my friend and I should spend our summer vacation in Norway or travel abroad. Norway is beautiful in the summer, but we don't want to skip on the opportunity to travel abroad when we have a vacation, as it is exotic. Since we have discussed the pros and cons extensively without reaching a clear conclusion, I decided to model the problem formally using an influence diagram. The goal is to determine which vacation option yields the highest expected utility given the uncertainties involved.

1.1 Decision Node

The decision node in the model is:

- **ChoiceOfVacation**: {Norway, International}

This represents the main choice between staying within Norway or traveling abroad.

1.2 Chance Variables

The model contains seven chance variables. Five of these are uncertain. Each variable is described below, together with its possible states.

- **FlightTicketPrice**: {Cheap, Expensive}
  Represents the uncertainty in flight prices. Prices depend heavily on season and availability. We know that flight prices for internation vacations are expensive during the summer. We can travel by car and train in Norway itself, making this variable a certain one as we know the value of the variable.

- **Weather**: {Sunny, Rainy}
  Weather significantly impacts enjoyment. Summer weather in Norway is highly variable, while international destinations may offer more stable conditions, but we cannot say for certain.

- **Language**: {Known, Unknown}
  In Norway, the language is always known, while international destinations can present a barrier. Language familiarity affects ease and enjoyment.

- **Crowd**: {High, Low}
  Popular tourist destinations vary in crowdedness. The high crowds reduce the enjoyment of the trip.

- **LivingExpenses**: {Cheap, Expensive}
  Accommodation and food prices is extremely high in Norway, whereas in a country like Spain it is much cheaper.

- **TravelDuration**: {Short, Long}
  International travel involves longer flights and transit times.

Longer travel duration increases stress and reduces ease. Travel time is generally sure shorter in Norway.

- **WillingFriend**: {Yes, No}
  Whether my friend is willing and available to travel. This uncertainty affects both enjoyment and ease. My friend is more excited for international travel. And I do not really care for a solo vacation, so I want her to come. So

1.3 Intermediate Value Nodes

To structure the model, I defined three intermediate factors that capture how the uncertain variables influence the final utility:

- **Cheap**: Indicates whether the overall trip is perceived as financially affordable.
- **Enjoyment**: Captures how fun or enjoyable the vacation is expected to be.
- **Ease**: Represents how stress-free and easy the vacation is expected to be.

Each of these nodes depends on a subset of the chance variables and the decision node. For example, *Cheap* depends on *FlightTicketPrice*, *LivingExpenses*, and the decision itself (Norway is more expensive than many international options). *Enjoyment* depends on weather, crowds, language familiarity, and whether my friend

is willing to join. *Ease* depends on travel duration, language, and crowdedness.

1.4 Graphical Structure

The diagram contains directed arcs from the chance variables to the intermediate nodes, and from the intermediate nodes to the utility node. The decision node is connected to intermediate value nodes.

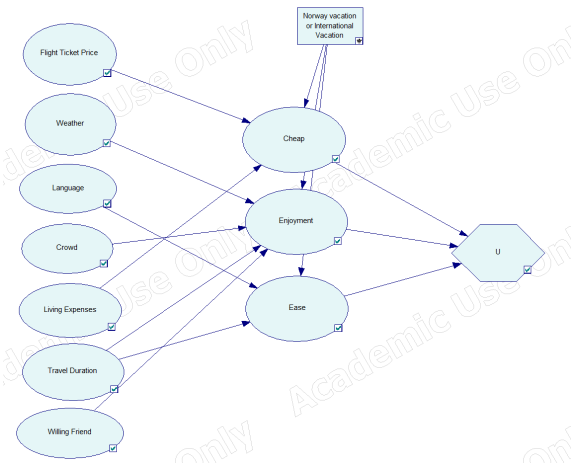

Figure 1: Diagram

1.5 Utility Function

The final utility is computed using a linear additive utility function:

$$U = 0.2 \cdot Cheap + 0.4 \cdot Enjoyment + 0.4 \cdot Ease.$$

The weights sum to 1 and reflect the relative importance of each factor. Affordability matters, but enjoyment and ease are way more important to me.

1.6 Quantification

I assigned probabilities subjectively based on typical expectations, following the calibration technique from Exercise 16.1.

Flight Ticket Price	Cheap				Expensive			
Living Expenses	Cheap		Expensive		Cheap		Expensive	
Norway vacation or International Vacation	Norway	International	Norway	International	Norway	International	Norway	International
Cheap_Vacation	0.8	0.9	0.5	0.6	0.4	0.5	0.2	0.3
Expensive_Vacation	0.2	0.1	0.5	0.4	0.6	0.5	0.8	0.7

Figure 2: Intermediate Value: Expenses

For example, if we choose an international vacation, and flight tickets are cheap along with living expenses, the trip will be cheaper as food and activities are also cheaper than Norway. Restaurants are more expensive in Norway. So I have given the probabilities based on logical and rational thinking.

Weather	Sunny															
Crowd	High								Low							
Willing Friend	Yes				No				Yes				No			
Travel Duration	Long		Short		Long		Short		Long		Short		Long		Short	
Norway vacatio...	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International
Fun	0.7	0.65	0.75	0.8	0.4	0.45	0.45	0.5	0.9	0.95	0.95	0.99	0.88	0.82	0.75	0.79
Not_Fun	0.3	0.35	0.25	0.2	0.6	0.55	0.55	0.5	0.1	0.05	0.05	0.01	0.12	0.18	0.25	0.21

Figure 3: Intermediate Value: Part 1 Enjoyment

The enjoyment category was harder to decide as it had many options. But the ideal situation was if the weather was sunny, my friend is willing, less crowded, and travel duration is short, the international vacation is ideal.

Weather	Rainy															
Crowd	High								Low							
Willing Friend	Yes				No				Yes				No			
Travel Duration	Long		Short		Long		Short		Long		Short		Long		Short	
Norway vacatio...	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International	Norway	International
Fun	0.25	0.4	0.3	0.45	0.03	0.15	0.1	0.2	0.38	0.42	0.69	0.75	0.2	0.23	0.21	0.24
Not_Fun	0.75	0.6	0.7	0.55	0.97	0.85	0.9	0.8	0.62	0.58	0.31	0.25	0.8	0.77	0.79	0.76

Figure 4: Intermediate Value: Part 2 Enjoyment

Language	Known				Unknown			
Travel Duration	Long		Short		Long		Short	
Norway vacatio...	Norway	International	Norway	International	Norway	International	Norway	International
Easy	0.9	0.79	0.95	0.98	0.99	0.85	1	0.88
Stressful	0.1	0.21	0.05	0.02	0.01	0.15	0	0.12

Figure 5: Intermediate Value :Ease

It is much easier and less stressful to take a vacation in Norway, as the language is known and most probably the travel duration will be short.

1.7 Purpose of the Model

The influence diagram was designed to incorporate the key uncertainties that affect the vacation outcome and to compute the expected utility of each alternative using GeNIe. Here is the result of the utility model when we choose between the two options:

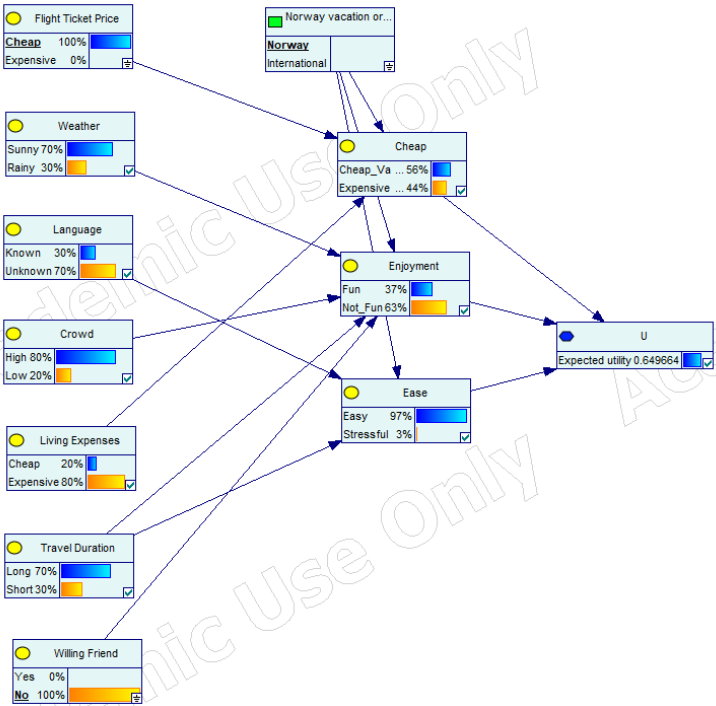

Figure 6: Utility result of Norway vacation

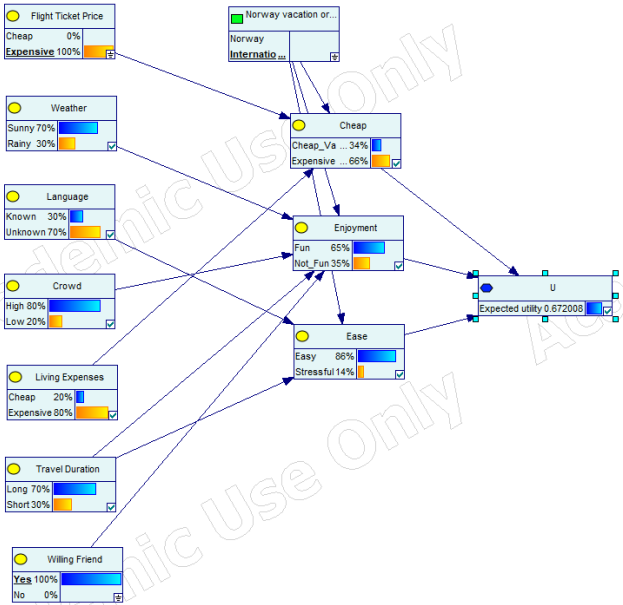

Figure 7: Utility result of International vacation

Looks like we're going international on vacation!